

Abdelaziz Omar

Backend Software Engineer

Faisal, Giza, Egypt • abdelaziz.omar405@gmail.com • +20 115 084 2586 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

SUMMARY

Backend Software Engineer specializing in Node.js, TypeScript, Express.js, and PostgreSQL, with hands-on experience building RESTful APIs, authentication systems, AI-powered applications, and scalable backend services. Developed and deployed multiple backend projects including healthcare, government-services, and real-time communication platforms, with experience in testing, API documentation, cloud deployment, and AI integration.

PROFESSIONAL EXPERIENCE

Stock Pilot

Jul 2026 - Present

Freelance | Full-Time

- Developed a multi-tenant inventory and sales management platform using Node.js, TypeScript, Express.js, PostgreSQL, and Sequelize.
- Implemented authentication, role-based authorization, warehouse management, inventory tracking, sales orders, and reporting workflows.
- Integrated Redis caching, background jobs, API documentation, testing, and deployment workflows.
- Applied modular backend architecture, validation, error handling, and audit logging practices.

Backend Developer

Aug 2024 - Sep 2024

[Robusta Studio](#) | Internship | Hybrid

New Cairo, Cairo, Egypt

- Contributed to a retail loyalty and commerce integration backend using Node.js, Express.js, PostgreSQL, Sequelize, and Redis.
- Implemented customer loyalty accounts, points and voucher workflows, order and refund webhooks, and transactional database updates.
- Applied JWT authentication, role-based authorization, request validation, rate limiting, audit logging, and centralized error handling.

Technology Intern

Jul 2025 - Aug 2025

[Agricultural Bank of Egypt](#) | Internship | On Site

Dokki, Giza, Egypt

- Completed technical assessments covering backend fundamentals, database concepts, secure software practices, and banking technology workflows.
- Gained exposure to core banking systems and how centralized platforms support accounts, transactions, financial operations, and internal services.
- Rotated across Web Development, IT, and Data Analysis teams to understand their workflows and collaboration within a banking environment.

MERN Stack Development Intern

Aug 2025 - Sep 2025

[National Telecommunication Institute \(NTI\)](#) | Summer Training | Hybrid

Giza, Egypt

- Completed practical full-stack training covering React.js, Node.js, Express.js, MongoDB, RESTful services, authentication, and deployment fundamentals.
- Built and integrated frontend and backend modules while applying CRUD operations, JWT authentication, validation, Git, and GitHub workflows.

Machine Learning Intern

Aug 2024 - Sep 2024

[National Telecommunication Institute \(NTI\)](#) | Summer Training | Hybrid

Giza, Egypt

- Studied supervised and unsupervised learning, data preprocessing, feature engineering, model evaluation, hyperparameter tuning, and neural-network fundamentals.
- Collaborated on an intrusion-detection classification project using the NSL-KDD dataset and compared multiple machine-learning models.

PROJECTS

Morgan's Hope Founder & Backend/AI Developer

Oct 2025 - Jul 2026

- Designed and developed an AI-assisted chest image screening platform, including backend architecture, REST APIs, authentication, database integration, medical-image workflows, and platform deployment.
- Implemented secure account flows, consent tracking, scan uploads, analysis history, structured result generation, PDF reports, hospital guidance, and a grounded medical assistant.
- Integrated a multi-model AI pipeline hosted as Docker-based services on Hugging Face Spaces and connected it to the Node.js backend through internal HTTP APIs.
- Technologies:** Node.js, TypeScript, Express.js, PostgreSQL, Sequelize, Supabase, JWT, OAuth 2.0, Docker, Vercel, Hugging Face Spaces, QStash, Redis, Swagger/OpenAPI.

Daleel  Backend Team Lead & AI Chatbot Developer

Feb 2026 - Jun 2026

- Led backend development for a government-services platform that organizes service requirements, procedures, documents, and practical guidance.
- Built modules for authentication, users, services, categories, comments, voting, settings, and user-submitted services with a pending-approval workflow.
- Developed a Groq-powered RAG-style chatbot grounded in a local government-services knowledge base and integrated Supabase persistence and Vercel deployment.
- **Technologies:** Node.js, Express.js, Supabase, PostgreSQL, REST APIs, Groq API, RAG-style Knowledge Grounding, Vercel, Git/GitHub.

PulseRooms  Real-Time Messaging Backend

Sep 2024

- Engineered a multi-room messaging backend with authenticated WebSocket connections and persistent message history in PostgreSQL.
- Implemented access and refresh tokens, protected room membership, role-based room management, request validation, and paginated message retrieval.
- Designed a relational schema for users, rooms, memberships, sessions, and messages with indexes for efficient room-based queries.
- **Technologies:** Node.js, Express.js, WebSockets, PostgreSQL, JWT, bcrypt.js.

EDUCATION

Culture & Science City, Faculty of Computer Science

Giza, Egypt

B.Sc. In Computer Science and Information Systems

2022-2026

- GPA: 3.04 / 4.00 | Graduation Project: Morgan's Hope (Excellent)

SKILLS & OTHER

- **Programming Languages:** JavaScript, TypeScript, Python, C#, C/C++.
- **Backend:** Node.js, Express.js, REST APIs, Sequelize ORM, JWT, OAuth 2.0, Passport.js, WebSockets.
- **Databases & Storage:** PostgreSQL, Supabase, MongoDB, MySQL, Redis, Upstash Redis.
- **Testing & Documentation:** Vitest, Supertest, Swagger/OpenAPI.
- **DevOps & Tools:** Git, GitHub, Docker, GitHub Actions, Vercel, Hugging Face Spaces, QStash, Sentry, Linux-based environments.
- **Languages:** Arabic - Native; English - Intermediate (B1).

VOLUNTEERING

Technical Instructor

Aug 2023 - Sep 2025

[Trainopia](#) | Internship | Hybrid

- Delivered introductory technical workshops and supported new participants throughout technical learning activities.
- Assisted learners in understanding programming fundamentals and practical software-development concepts.